

PROTOTYPE:

EVERYTHING = NOWHERE = NOW

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CART 360: Tangible Media & Physical Computing

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GitHub Repository

<https://github.com/Jimmy-Le/SonicScrewdriver>

Project Description

Our artifact was originally inspired by the “Sonic Screwdriver” from the Doctor Who series. This multi-purpose tool can open doors, manipulate electronics, send messages, and more while emitting light and sounds. We have instead decided to make it more interactive for users by making 2 separate “screwdrivers” (without the original functionalities) that behave differently based on the distance between them, with each having its own personality (tone and light).

The goal of our project is to help bring people together in a world where technology isolates us, especially when our main method of communication is through text messages and social media. Our artifact is intended for friends, coworkers, or families in the same building who wish to make time to hang out, as the “screwdrivers” will emit slow and sad beeping noises when they are too far away, and more lively noises and lights as they approach in real time. This encourages users to get away from their screens, go search for each other, and play around with the artifact in their hand.

The beeping tone is also affected by the user’s movement. As they move the screwdriver around, they get feedback based on the rotation of the device, which changes the frequency of the buzzer. Thus, the users get empowered to move and play around each other to create a fun little symphony.

Development Process

As we are both inexperienced with electronics, our main concern was seeing if we could create the prototype within less than a week. We decided to try to do some personal research before our first meeting to make the process easier. After talking to the professor, we concluded that we needed to learn more about RSSI Signals and hosting a Wifi network on a Raspberry Pi Pico W.

On our initial meeting day, we both came up with a little Micropython code to set up the Access Point, and try to get the RSSI signal. This was a success. We managed to have a Pi host a network, and the other to connect and calculate the RSSI; however, the host cannot receive that information back. We tried several solutions online, yet none of them worked, or there were bugs that we didn't know how to solve at the time. We spent over 4 hours trying to figure it out, but in the end, we gave up on this and asked the professor for help. Meanwhile, we coded up the different features (LED, Buzzer, and Gyro) using the RSSI of the Access Point on the connecting Pi. We also experimented a bit with the audio and speaker module, but it didn't work as well, so we went back to using Buzzers as our main audio source.

After our next meeting, the professor recommended that we use sockets to send UDP messages to the host. We initially tried this method, but there were issues where it would get stuck in an infinite busy wait loop. It turns out that when we close the program, the sockets weren't closing, causing the connection issues. After this was solved, we tried copying the code to the host Pi using the received RSSI. While it works, the issue is that the messages stop getting received after a certain range or abrupt movement. We decided to adapt our code to change behavior based on a smaller range of movements to help display the functionalities for the demo.

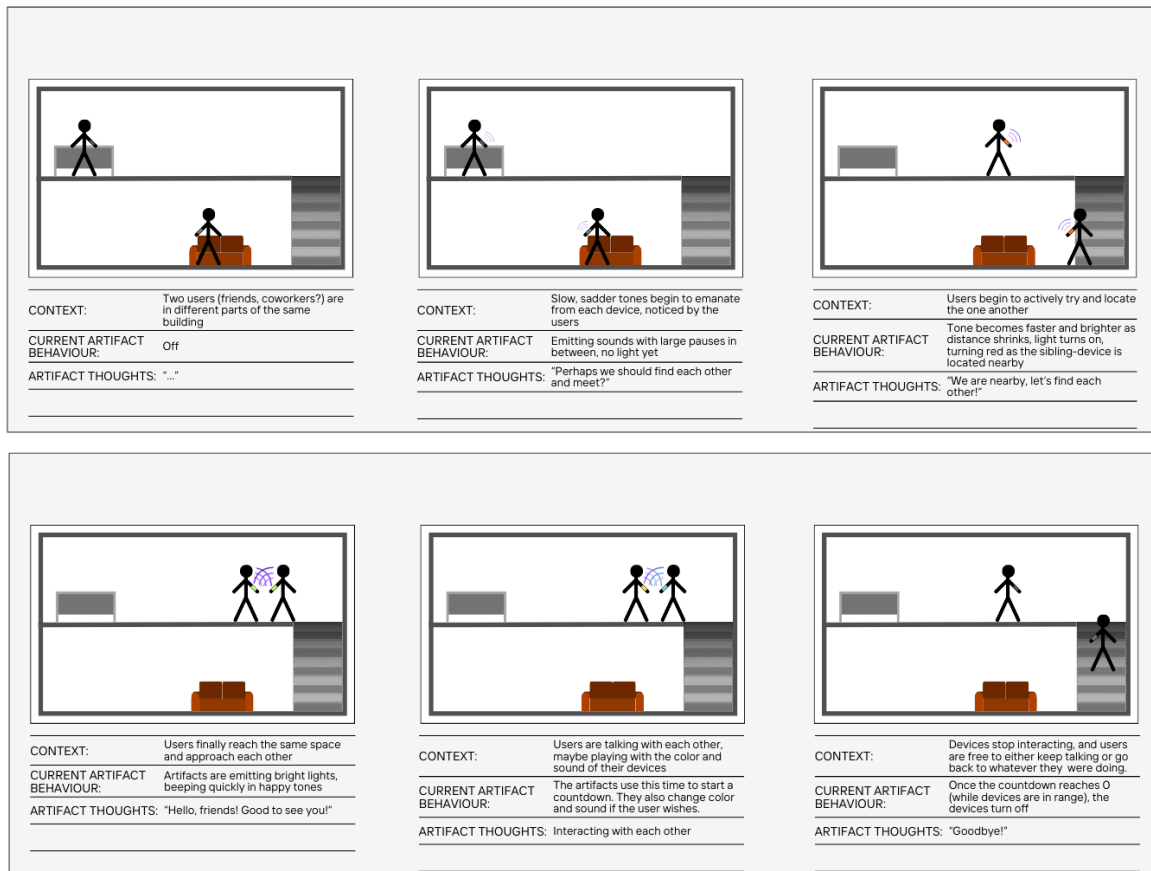
In the future, we could probably try increasing the range with antennas to make it more fun to run around a building searching for each other, but based on time and budget limitations, we might proceed in a different direction.

Sensor Evaluation & Affordances

MPU 6050 (Gyro) - Our project uses this Gyroscopic sensor to figure out whether the board is rotated from its horizontal position. Tilting the board produces numerical rotation data that we map to changes in the buzzer's pitch, creating an interaction between the user and board that can influence audio output through rotation. The two different boards each respond with a different frequency change. This gives our boards separate personalities, affording the user a varied experience and solidifying the 'pet-like' association of this project.

RSSI (Receive Signal Strength Indicator) from built-in sensors - We use the built-in RSSI sensor as the base of all the digital parts of the project. The values from the signal strength are used to estimate the distance between the two devices. Bringing them closer makes the signal stronger, which causes the mapped data to make the buzzer make sounds at a faster rate, as well as the LEDs to light up (amount & color depending on distance). This solidifies the relationship between each board, with them responding to each other both with audio and visuals. This sensor affords a spatial awareness to the devices, allowing users to influence them by physically moving them either closer or further apart. On the other hand, the boards influence the user by creating an incentive / demand to come closer or get further apart. This all goes to make the devices even more responsive and feel more socially alive.

Interaction Design Strategies



Similar Projects

[Totem Compass](#)

A necklace that syncs with up to 4 other Totem Compasses. This product lets users find their friends by tilting their Totem horizontally, which will then display lights that indicate the direction of their friends in real time without service or wifi. It also has a "Vibe" mode, which controls the

LED based on the surrounding music. The intended environment for this product is for concerts or big social events, where one might get easily separated from their group.

Our product is different because it is more personal, as it is designed to be between 2 people. It also encourages the users to seek each other out and play, while the Totem Compass is used as a tool to reconvene when they are lost. As such, our artifact will be more playful in nature and used to help form meaningful bonds.



[Buchla Lightning](#)

The Buchla Lightning is a tool/instrument created by Don Buchla that comes with a box and 2 wands, and takes the movement and velocity of the wands as an input for musical sounds (MIDI Controllers). It can also come with foot pedals that can control the type of instrument or sound it is playing. For example, you can “play” the drums with the wands by smacking them in the air, with the velocity affecting the intensity of the sound output.

An example of a performance using the Buchla Lightning:

<https://www.youtube.com/watch?v=HSf8sYHWBsQ>

The primary purpose of our product is to bring people together and start a conversation. The Buchla Lightning is an instrument that is mainly played by a single person with the goal of creating music. They are similar in the way that they both utilize a “wand” that creates noise based on their movement in the physical space, but our screwdrivers help to connect people.

Bond Touch

Bond Touch is a pair of bracelets marketed for couples to form a physical connection while miles apart. They allow each other to send a vibration or display a color on their partner’s bracelet to let them know that they are thinking about them. This is useful for long-distance relationships, for people who work a lot, and some even use it as a medical alert to notify their partner about a potential episode.

Our product can’t reach too far, so it will be mainly for people to reconnect with one another within the same building or household. It will focus more on being in the moment with one another, promoting physical movements, which the bond touch can’t replicate as well.



Task Allocation

Tasks	Person	Date
Visit the Sensor Lab for ideas and help	Jimmy	FEB 25
Research RSSI	Jimmy	FEB 27
Research AP / Wifi hotspot	Sean	FEB 27
Experimentation Assemble Electronics Create Hotspots Access RSSI Setup LED Setup Buzzer Setup Gyro Debugging	Jimmy Sean	FEB 28

Experimentation Implement UDP broadcast Adapt the LED, Buzzer, and Gyro to the Host Pi Reduce the distance for customization Debugging	Jimmy Sean	MAR 5
Project Description 3 Similar Product Research	Jimmy	MAR 5 - 9
Non-Technical Sensor Description Storyboarding	Sean	MAR 5 - 9